

Exploring Radar Data Representations in Autonomous Driving: A Comprehensive Review

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Abstract—With the rapid advancements of sensor technology and deep learning, autonomous driving systems are providing safe and efficient access to intelligent vehicles as well as intelligent transportation. Among these equipped sensors, the radar sensor plays a crucial role in providing robust perception information in diverse environmental conditions. This review focuses on exploring different radar data representations utilized in autonomous driving systems. Firstly, we introduce the capabilities and limitations of the radar sensor by examining the working principles of radar perception and signal processing of radar measurements. Then, we delve into the generation process of five radar representations, including the ADC signal, radar tensor, point cloud, grid map, and micro-Doppler signature. For each radar representation, we examine the related datasets, methods, advantages and limitations. Furthermore, we discuss the challenges faced in these data representations and propose potential research directions. Above all, this comprehensive review offers an in-depth insight

into how these representations enhance autonomous system capabilities, providing guidance for radar perception researchers. To facilitate retrieval and comparison of different data representations, datasets and methods, we provide an interactive website at <https://radar-camera-fusion.github.io/radar>.

Index Terms—Radar perception, autonomous driving, data representations, intelligent vehicles, intelligent transportation.

I. INTRODUCTION

NOWADAYS, the automotive industry has witnessed significant advancements in autonomous driving technologies, revolutionizing intelligent vehicles and intelligent transportation. With the aim of creating safer and more efficient roadways, one crucial aspect is the development of reliable sensor perception for autonomous driving systems. Compared to LiDARs and cameras, the fundamental advantage of radar sensors lies in their capabilities in range, velocity and angle measurements [1], [2]. Additionally, radars have robust sensing capabilities and exhibit superior effectiveness under adverse lighting and weather conditions [3], [4]. Moreover, radar sensors can even perceive objects behind walls, providing vehicles with early warning of potential obstacles or hazards [5], [6]. All of the above characteristics make radar an indispensable component in ensuring the reliability and superiority of autonomous driving systems, especially in scenarios where other sensor inputs might be limited or compromised.

As depicted in Fig. 1, radar sensors have extensive applications in autonomous driving, serving various purposes in the perception of the surrounding environment. Radars mounted on the roof, front and angular directions of an intelligent vehicle are utilized to detect objects in the path and surroundings, providing essential information for path planning and adaptive cruise control [7], [8], [9]. The radar inside the cabin is leveraged to monitor occupant vital signs, left-behind children and driver behaviors (e.g., drowsiness, dangerous maneuvers, health emergencies) in a privacy-protected manner, allowing the autonomous driving system to make alerts as well as safety measures [10], [11], [12], [13]. Furthermore, radar systems deployed alongside roadways can measure the speed of passing vehicles and estimate traffic flow. These traffic data are valuable for traffic optimization, congestion management and intelligent transportation systems, leading to improved road safety and smoother traffic operations [14], [15], [16].

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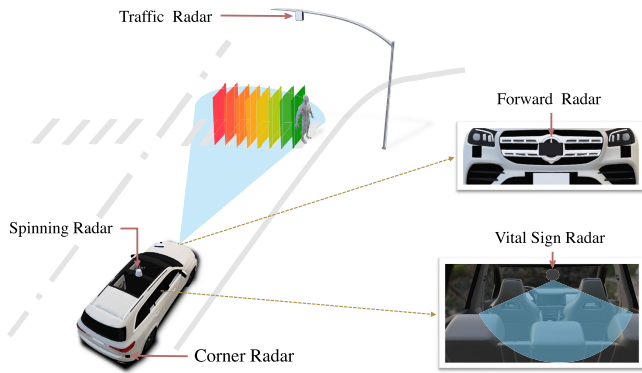


Fig. 1. Radar perception in autonomous driving. (a) Radars on the vehicle are employed to detect objects in the path and surroundings. (b) Radars inside the cabin are leveraged to monitor occupant vital signs and behaviors. (c) Radars alongside roadways are utilized to measure the speed of passing vehicles and estimate traffic flow.

As far as radar data is concerned, various application scenarios and radar types provide different data representations. Specifically, each representation has distinct characteristics and challenges, involving different data processing methods and network architectures. Moreover, each representation aims to extract needed information from raw radar measurements and convert it into a suitable format for downstream perception tasks. With limited focus on radar data representations in existing reviews, it is challenging for researchers to comprehensively understand this emerging autonomous driving field. This review attempts to narrow this gap by exploring five radar representations (i.e., ADC signal, radar tensor, point cloud, grid map, micro-Doppler signature) leveraged in autonomous driving and their impact on perception tasks. Specifically, we provide guidance for researchers to differentiate between each representation, understand the advantages and limitations of each, and select the appropriate representation and algorithm for the specific task. Our review offers the following key contributions:

- To the best of our knowledge, this is the first review that explores different data representations for radar perception in autonomous driving.
- We offer an up-to-date (2019-2024) overview of radar-based datasets and algorithms, providing in-depth research on their advantages and limitations.
- We analyze the significant challenges and open questions related to these data representations, and propose potential research directions for further investigation.
- We develop an interactive and regularly updated website that facilitates easier retrieval and comparison of the datasets and methods.

The remainder of this review is structured in Fig. 2 and is described as follows: Section II provides an overview of radar working principles and signal processing techniques, serving as the foundation for generating various radar data representations. Section III presents an in-depth examination on datasets and methods for each radar data representation. Additionally, this section discusses the advantages and limitations of each radar data representation, assisting readers in selecting the most suitable approach for their research. Section IV delves into challenges and potential directions in

research and development for radar perception in autonomous driving, guiding readers to focus on current hot topics and explore feasible solutions. Lastly, Section V summarizes our study and presents an outlook for future works, inspiring researchers to make further strides in radar perception.

II. RADAR PERCEPTION

In this section, we provide an overview of the underlying principles of radar technology and how it operates within the autonomous driving system. We delve into the physics behind radar sensing, including the transmission and reception of radio waves, as well as the mechanisms for extracting valuable information from the returned signals. Understanding these principles is crucial for comprehending the capabilities of radar perception and the generation process of various radar data representations.

A. Working Principles

Radar sensors play crucial roles in enabling autonomous systems to sense vehicle occupants, vehicle surroundings, and the traffic environment. The term “radar” stands for “Radio Detection and Ranging”, which emits millimeter waves that bounce off objects and return to the sensor. This operation provides information about the object’s location, relative velocity, and internal characteristics [17]. In the following, we introduce the basic radar working pipeline and sensor characteristics derived from these principles.

1) *Basic Pipeline*: Fig. 3 demonstrates a basic radar system that comprises various analog components, including a synthesizer, a Power Amplifier (PA), transmitters (TX antennas), receivers (RX antennas), a Low Noise Amplifier (LNA), a mixer and a Low Pass Filter (LPF), as well as digital components such as an Analog-to-Digital Converter (ADC), a Digital Signal Processor (DSP) and a MicroController Unit (MCU). When the radar starts working, first, the synthesizer generates a linear frequency-modulated pulse called a “chirp”, which is amplified in power and transmitted by the TX antenna. Second, the RX antenna captures the reflected chirp of the target that is amplified with low noise. Third, by combining the RX and TX signals, the mixer produces an Intermediate Frequency (IF) signal, which is then converted into digital values via the ADC. Generally, a radar system contains multiple TX and RX antennas, resulting in multiple IF signals. Information about the target object, such as range, Doppler velocity and azimuth angle, is contained in these IF signals, which can be separated by the DSP using embedding different signal processing algorithms [18]. Based on extracted target information from the DSP and dynamic information from the radar, the MCU serves as a computer to evaluate system requirements and make informed decisions.

Utilizing the Time of Flight (TOF) principle, radar sensors determine the range of an object by calculating the time difference between the transmitted and reflected signals. Additionally, based on the Doppler principle, radar sensors can measure the relative Doppler velocity of a target by analyzing the frequency difference between the emitted and received waves, capturing any relative movement between

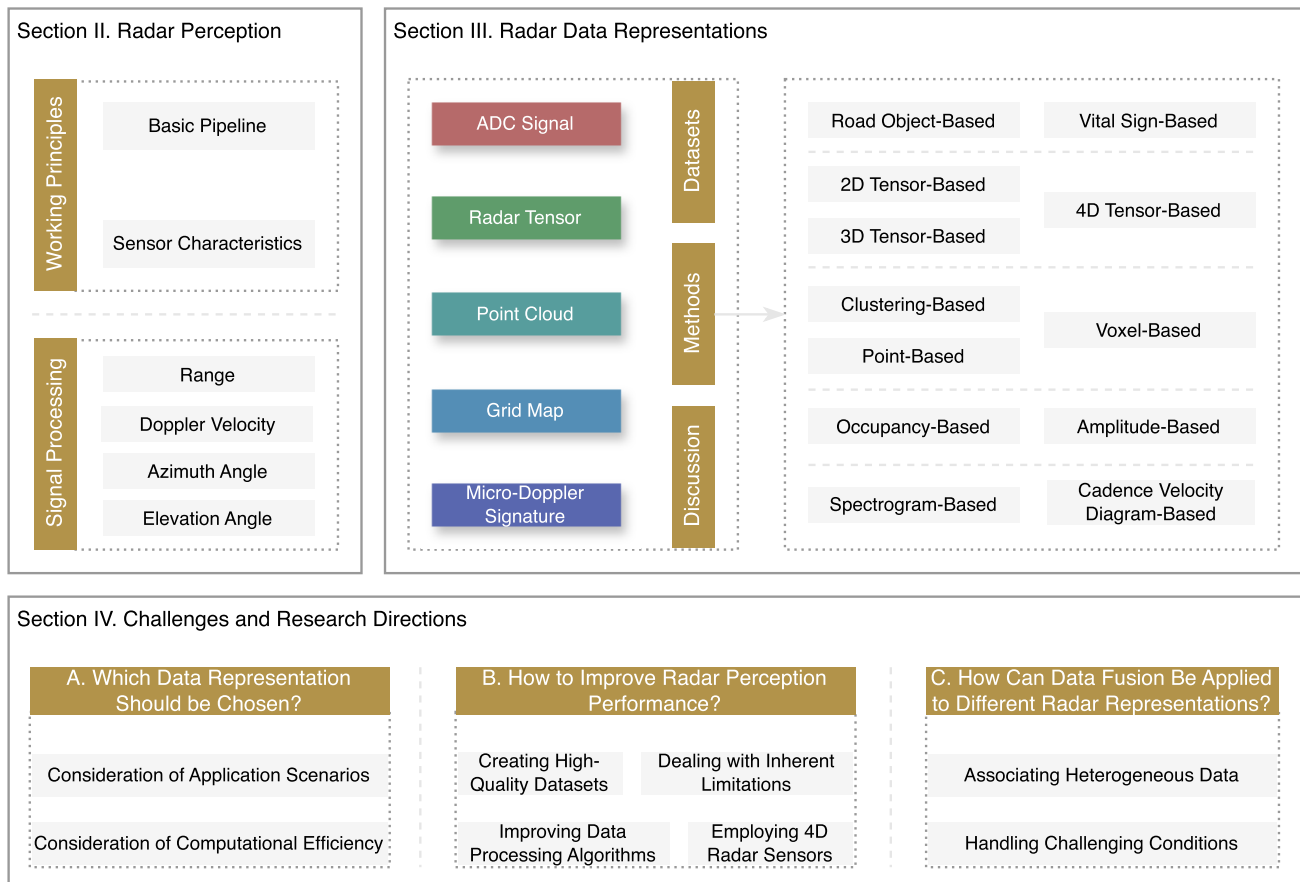


Fig. 2. Overview of this review. Section II provides an overview of radar perception, including its working principles and signal processing techniques. Section III presents an in-depth examination on datasets and methods of different radar data representations. Section IV discusses the challenges and potential directions for research and development for radar perception in autonomous driving.

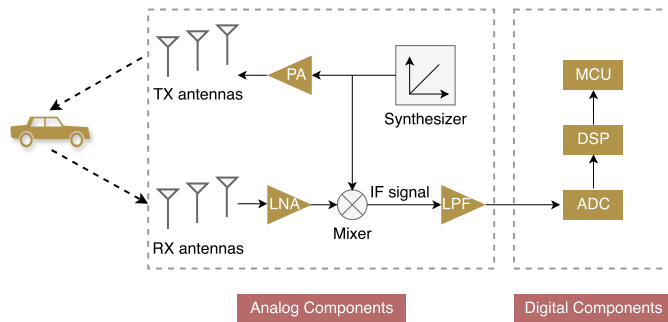


Fig. 3. Overview of radar working pipeline.

the radar and the target. To determine the azimuth angle, an array signal processing method is employed, which involves calculating the phase difference between the chirps reflected from parallel receivers. However, traditional 3D radar sensors with receivers arranged in a 2D manner only provide detection in horizontal coordinates, lacking vertical height information. Recent advancements in radar technologies have led to the development of 4D radar sensors, incorporating antennas arranged both horizontally and vertically [19], [20]. This configuration allows for the measurement of the elevation angle, thereby enabling the capture of elevation information in addition to the range, Doppler velocity, and azimuth angle measurements.

In pipeline systems, the number of steps is a critical factor that directly impacts complexity, performance, modularity, and maintainability. A larger number of steps enhances modularity by breaking down complex tasks into smaller, more manageable components, which improves maintainability and facilitates debugging. However, an excessive number of steps can increase computational overhead, introduce latency, and complicate system integration. These trade-offs must be carefully balanced based on the specific requirements of the application.

2) *Sensor Characteristics*: With the basic capabilities of measuring range, Doppler velocity, azimuth angle and elevation angle, radar can determine the location of obstacles, allowing vehicles to make informed decisions and navigate safely. Unlike other light-wave-based sensors, radar sensors emit radio waves at longer wavelengths. This characteristic allows radar waves to penetrate fog, rain, snow, smoke, and dust [21]. Consequently, radar systems can reliably detect and measure distances to objects even in severe weather conditions, making them highly dependable in a wide range of real-world scenarios. Moreover, radar waves can penetrate certain materials (e.g., walls, vegetation) and reflect off the hidden objects, enabling the sensor to detect objects situated around corners or obstructed by other obstacles [5]. This unique characteristic enhances the detection capabilities of the system,

TABLE I
SUMMARY OF RADAR PARAMETER ESTIMATION [17]

Parameter	Estimation	Resolution	Notes
Range	$R = \frac{cf_b}{2S}$ (1)	$R_{res} = \frac{c}{2B}$ (2)	c is the speed of the light ($3 \times 10^8 m/s$). f_b is the instantaneous frequency difference at the mixers from TX-chirp and RX-chirp. S is the slope of a chirp. B refers to the bandwidth.
Doppler Velocity	$V = \frac{\lambda \Delta \phi}{4\pi T_c}$ (3)	$V_{res} = \frac{\lambda}{2T_f}$ (4)	λ is the wavelength of the transmitted signal. $\Delta \phi$ is a shift in phase. T_c is the duration between the chirps. T_f is cycle time of a frame.
Azimuth Angle	$\theta = \sin^{-1}(\frac{\lambda \Delta \phi}{2\pi l_a})$ (5)	$\theta_{res} = \frac{\lambda}{N_a l_a \cos(\theta)}$ (6)	l_a is the length between RX antennas in azimuth. N_a is the number of RX antennas in azimuth.
Elevation Angle	$\theta = \sin^{-1}(\frac{\lambda \Delta \phi}{2\pi l_e})$ (7)	$\theta_{res} = \frac{\lambda}{N_e l_e \cos(\theta)}$ (8)	l_e is the length between RX antennas in elevation. N_e is the number of RX antennas in elevation.

particularly in complex urban environments where numerous obstacles may impede direct line of sight.

While radar sensors offer numerous advantages, it is essential to consider their inherent limitations. They exhibit limited angular resolution, making it challenging to differentiate between closely located objects. Additionally, sparse point clouds generated by radars, with only a few points on pedestrians and a small number on cars, are insufficient for accurately outlining object contours and extracting detailed geometric information [22], [23]. While radar measurements provide the radial velocity component, they lack information about tangential velocity. This limitation makes it difficult to accurately estimate the complete motion of objects in dynamic scenes [24], [25]. More importantly, radar data is susceptible to noise from various sources, including multi-path interference, electrical interference, and equipment imperfections [26], [27], [28]. This noise impacts the precision and reliability of radar measurements, potentially leading to false detections.

B. Signal Processing

In this section, we review the radar signal processing for radar parameters, including range, Doppler velocity, azimuth angle and elevation angle. Subsequently, Radar Cross-Section (RCS) measurement is analyzed to approximate the target's size, shape and material composition, thereby characterizing its reflection properties. To filter out clutter during the radar signal processing stage, we examine the workflow of Constant False Alarm Rate (CFAR) processing and explore representative CFAR processors.

1) *Parameter Estimation:* Frequency-Modulated Continuous Wave (FMCW) is a special millimeter wave technology employed in autonomous driving that continuously transmits frequency-modulated signals to measure the attributes of objects. As is described in Fig. 4(a), the frequency of the chirp emitted by the TX antenna increases linearly over time, leading to the corresponding frequency of the chirp reflected by the RX antenna. The IF signal containing information on objects is

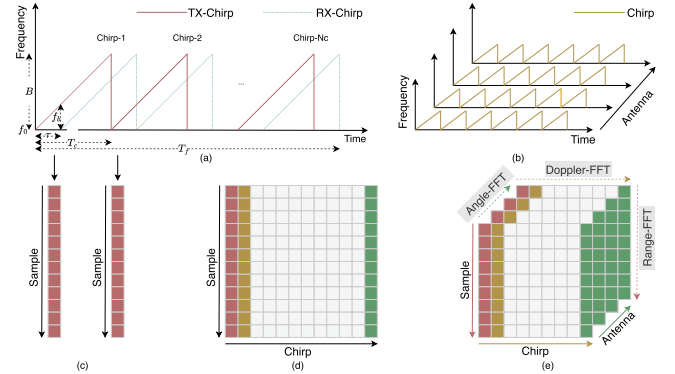


Fig. 4. Pipeline of radar signal processing. (a) Frequency of chirps emitted by a TX antenna and received by an RX antenna. (b) Frequency of chirps received by multiple RX antennas. (c) Sampling performed on each chirp. (d) Sample-Chirp map generated from sampling on all chirps. (e) Simple-Chirp-Antenna tensor generated from Sample-Chirp maps based on multiple RX antennas.

determined by calculating the difference in the instantaneous frequency of the TX-chirp and RX-chirp, expressed by the variable f_b . In the following, we introduce the estimation of each parameter and summarize the mathematical formulation in Table I.

a) *Range:* As depicted in Fig. 4(c), sampling is initially conducted among each chirp signal. Then, the Fast Fourier Transform (FFT) is performed to produce a spectrum with different peaks, representing different objects at different ranges. Given that FFT operation is utilized to obtain range values, it is also called “Range-FFT”. From Equation 1, we know that the range of a target is determined by the difference in the frequency and the slope of a chirp. Range resolution refers to the capability of distinguishing and resolving two objects that are located very close together along the range dimension. According to Fourier transform theory, the range resolution can be improved by extending the IF signal [29]. Extending the IF signal requires increasing the signal bandwidth, which

results in range resolution being proportional to the signal bandwidth (B), as shown in Equation 2.

b) Doppler Velocity: To measure the Doppler velocity of an object using radar, a common technique involves transmitting two chirps that are separated by a time interval denoted as T_c . For each chirp, the spectrum after Range-FFT peaks at the same position but with different phases. The difference of the measured phase $\Delta\phi$ contains the velocity of an object, as noted in Equation 3. For multiple objects at the same range, it is difficult to identify their velocities as the peaks are located at the same position. Thus, a sequence of signals consisting of N_c chirps is employed, formulating a Simple-Chirp map described in Fig. 4(d). Doppler-FFT is then performed on the phasors generated by Range-FFT to separate objects since the phase differences between consecutive chirps are different. Velocity resolution is the minimum difference in velocity at which a radar can distinguish between two targets at the same range. As expressed in Equation 4, velocity resolution is inversely proportional to the cycle time of a frame denoted as T_f .

c) Azimuth Angle: Estimating the azimuth angle is accomplished using at least two RX antennas separated by l_a . The resulting difference in distances from the target to each RX antenna causes a phase change in the FFT spectrum, from which the Direction of Arrival (DoA) can be obtained, as outlined in Equation 5. To measure the azimuth angle of multiple objects located at the same range and moving at the same velocity, multiple RX antennas are needed, as shown in Fig. 4(b). In this way, the radar signal is described in a Simple-Chirp-Antenna dimension, as illustrated in Fig. 4(e). Then, an Angle-FFT is performed on the phasor sequences corresponding to the peaks after 2D-FFT (Range-FFT and Doppler-FFT) to resolve azimuth angles. Azimuth angle resolution θ_{res} is the minimum angle separation for two objects appearing as separated peaks in the spectrum after Angle-FFT. It can be justified from Equation 6 that the azimuth angle resolution is maximum when measured perpendicular to the radar system's axis ($\theta = 0$). Additionally, it is necessary to increase the number of RX antennas to enhance azimuth angle resolution.

d) Elevation Angle: Similar to azimuth angle estimation, the measurement of elevation angle requires a minimum of two RX antennas, each separated by a specific length, denoted as l_e . The mathematical formulation that describes this azimuth angle estimation is expressed in Equation 7. Additionally, the resolution of elevation angle can be calculated using Equation 8, which can be improved by increasing the number of RX antennas.

2) RCS Measurement: RCS denotes the ability of an object to reflect a radar signal, and a higher RCS value corresponds to an increased likelihood of detection [30]. The value of RCS is expressed as an area in m^2 [31]. However, this value does not simply represent the surface area of the object being detected, but depends on multiple factors, including the target's material, physical geometry, and exterior features of the target, as well as the direction and frequency of the illuminating radar.

In terms of mathematical calculations, RCS is a metric that quantifies the ratio between the scattered density in the direction of the radar and the power density intercepted by

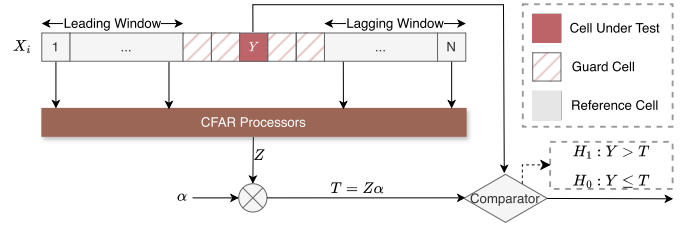


Fig. 5. Overview of CFAR processing. Set of $X_1, X_2, \dots, X_N$ represents the detection window, Y is the value of CUT, Z represents the clutter background level of the CUT, T is the detection threshold, α denotes a scaling factor, H_1 declares that an object is located within the CUT, H_0 indicates that there is no object in the current CUT.

the object. Since the power is distributed over a sphere, only a small part of this ($4\pi r^2$) can be received by the radar. Hence, the expression for RCS takes the form in Equation 9:

$$\sigma = \frac{4\pi r^2 S_r}{S_t}, \quad (9)$$

where r refers to the range between the radar and the target, S_r is the scattered power density at the radar, and S_t represents the incident power density the target intercepts.

3) CFAR Processing: The radar sensors not only receive the reflected signals from the objects of interest, but also encounter the internal receiver noise and external interfering signals. Signals generated by these unwanted sources are commonly called clutter. Traditional methods like removing signals with zero Doppler and fixing signal thresholds have drawbacks that lead to false alarms. Consequently, dynamic thresholding, which involves using adaptive threshold values, is crucial in mitigating false alarms and spurious radar detections attributable to noise signals. Compared to fixed thresholds, varied thresholds are involved in reducing false alarms and erroneous radar detections caused by noise. CFAR is the most commonly used method of dynamic thresholding [32], [33], facilitating the radar system to autonomously adapt its sensitivity threshold in response to variations in the amplitude of external interference, thus ensuring a consistent level of false alarm probability.

Fig. 5 describes the overview processing flow of CFAR processors. Before feeding the data into the CFAR processor, a square law detector is applied to convert the real and imaginary parts of the radar data into a real-valued square of its power [34]. The radar signal then runs through a sliding window comprising reference cells, guard cells, and Cell Under Test (CUT). In the estimation of the clutter background level, the utilization of neighboring reference cells surrounding the CUT is justified owing to the spatial-temporal correlation exhibited by radar echoes. Then, employing various detectors, the clutter background level of the CUT (referred to as Z) can be assessed by utilizing samples within the reference window. The threshold value T is calculated by multiplying Z by a scaling factor α . If the value of the CUT (labeled as Y) exceeds the threshold value T , the comparator will declare that an object is located within the CUT (expressed as H_1). Otherwise, there is no object in the current CUT (expressed as H_0).

A reasonable selection of processors for the clutter statistical model in the CFAR clutter window can significantly mitigate

the issues posed by intricate scenarios, such as uniform clutter, multiple targets, and clutter edges. As a result of the multitude of techniques available for estimating clutter within the CFAR framework, a variety of CFAR methods have been proposed. These methods can be broadly classified into three types:

a) *Mean-Level Processors*: The first type of processor comprises mean-level estimators, including Cell Averaging CFAR (CA-CFAR) [35], Greatest-of CFAR (GO-CFAR) [36] and Smallest-of CFAR (SO-CFAR) [37]. These processors obtain the estimated clutter power level by taking the average value, maximum value and minimum value of the reference cells, respectively. However, the detection mechanism in mean-level processors is not optimized for multiple objects. Other objects within the reference window distort the noise estimation and lead to an increased threshold value, thereby causing potential target detections to be missed.

b) *Sorting-Based Processors*: The second category of the processor is exemplified by Ordered Statistics CFAR (OS-CFAR) [32], which sorts the reference cells in ascending order and selects the k^{th} value as the estimated clutter power level. After that, numerous methods based on ordered statistics have emerged, called OS-like methods, such as Censored Mean Level Detector CFAR (CMLD-CFAR) [38], and Trimmed Mean CFAR (TM-CFAR) [33]. However, since the sorting-based processors retain only one ordered reference sample, they rely heavily on prior knowledge about the number and distribution of interference objects.

c) *Neural Network-Based Processors*: Recently, with the rapid development of machine learning and deep learning, traditional CFAR processors have evolved into neural network-based methods. A representative example of this method is DBSCAN-CFAR [39], which combines artificial neural network and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) as a clustering algorithm [40]. Unlike conventional methods, DBSCAN-CFAR is able to learn the underlying relationship between normal data and outliers, even in the absence of labeled data or predetermined information regarding the number of clusters. In simulations involving varying object quantities, shape parameters, and false alarm probabilities, DBSCAN-CFAR exhibits superiority and robustness compared to traditional processors like CA-CFAR, SO-CFAR, GO-CFAR, OS-CFAR, and CMLD-CFAR [39]. However, to achieve the advantages mentioned above, DBSCAN-CFAR suffers from a higher computational burden and time consumption when compared to other processors.

III. RADAR DATA REPRESENTATIONS

In this section, we explore what are radar data representations in autonomous driving and how these representations enhance autonomous driving capabilities. Firstly, we introduce the generation progress of different radar data representations and their basic characteristics. Subsequently, for each data representation, we explore the related datasets and representative methods, and discuss the benefits and limitations associated with each representation, thus exploring their capabilities in various perception tasks such as classification, localization,

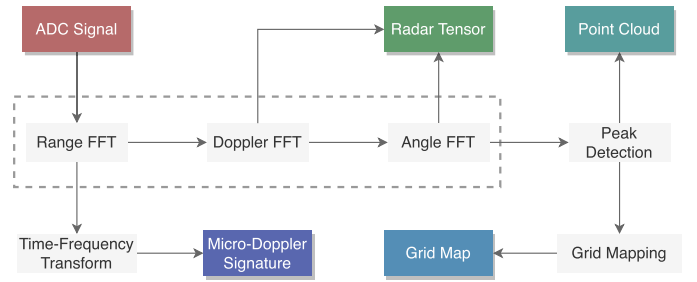


Fig. 6. Generation progress of five radar data representations (i.e., ADC signal, radar tensor, point cloud, grid map, and micro-Doppler signature).

detection, and tracking. In addition, we also outline datasets for radar perception in autonomous driving in Table II.

Fig. 6 illustrates the generation process of five different radar data representations. Initially, the raw output from the radar sensor, known as the **ADC signal**, which is a digital representation of the analog waveforms received by the radar antenna. The ADC signal is a complex and high-dimensional dataset that contains a mixture of target returns, noise, and clutter, which is hardly interpretable by human observers. Thus, researchers utilize 3D FFTs across the sample, chirp, and antenna dimensions to transform the ADC signal into an image-like representation called the **radar tensor**. The radar tensor organizes the data into a multi-dimensional array where each point corresponds to a specific range, Doppler shift, and angle, making it easier to visualize and analyze the radar returns. Subsequently, peak detection is employed on the radar tensor to eliminate clutter, resulting in a sparse, point-like representation referred to as the **point cloud**. Each point in the cloud represents a potential object in the environment, with its coordinates corresponding to the object's range, azimuth, and elevation. By accumulating point clouds over a specific duration and applying grid mapping methods, the **grid map** representation is generated for the purpose of identifying static objects. Each cell in the grid map contains information about the occupancy or probability of occupancy of that particular location in the environment. Moreover, some researchers perform a Time-Frequency transform following the Range-FFT to extract the **micro-Doppler signature**, which is utilized for recognizing objects characterized by micro-motion features. The micro-Doppler signature is vital for the recognition of objects with discernible micro-motion patterns, enhancing the capabilities of radar systems in various surveillance and identification scenarios.

A. ADC Signal

When an analog signal is sampled and quantized by an ADC, it produces a sequential data stream called the ADC signal. As the raw data produced by radar sensors, ADC signals retain all information from the detections, which is highly valuable for deep learning applications. At this stage, the signal lacks spatial coherence among its values, as all information is confined to the time domain [48]. To be represented in a more structured format, the ADC signal is typically converted to a 3D Sample-Chirp-Antenna (SCA) tensor, as illustrated in Fig. 4(e).

TABLE II
DATASETS FOR RADAR PERCEPTION IN AUTONOMOUS DRIVING

Dataset	Representations	Year	Tasks	Sensors	Scenarios
RaDiCaL [41]	ADC Signal	2021	Object Detection	Radar (TI IWR1443), RGB-D Camera	Indoor (people, static clutter), Roads (urban, rural, highway, various traffic scenarios)
RADial [42]	ADC Signal, Radar Tensor, Point Cloud	2021	Object Detection, Semantic Segmentation	Radar (high-definition), Cameras, LiDAR	Roads (urban, highway, rural)
CARRADA [43]	Radar Tensor	2020	Detection, Semantic Segmentation, Object Tracking	Radar (TI AWR1843), RGB-D Camera, LiDAR	Roads (urban, highway, intersection scenarios)
Zendar [44]	Radar Tensor, Point Cloud	2020	Object Detection, Mapping, Localization	Radar (synthetic aperture), Camera, LiDAR	Roads (diverse urban driving environments)
RADIATE [45]	Radar Tensor	2020	Object Detection	Radar (Navtech CTS350-X), Camera	Roads (wet, snowy, foggy, rainy, nighttime, urban, highway)
MulRan [46]	Radar Tensor	2020	Place Recognition	Radar (Navtech CIR204-H), Cameras, LiDAR	Roads (city, highway, intersection, crosswalks, parks, recreational areas, tunnels, bridges)
Oxford Radar RobotCar [47]	Radar Tensor, Grid Map	2020	Object Detection, Odometry	Radar (Navtech CTS350-X), Camera, LiDAR, GPS, INS	Roads (urban, highway, rural, industrial area, residential area, roundabout, intersection)
SCORP [48]	Radar Tensor	2020	Semantic Segmentation	Radar (76 GHz), Camera	Roads (parking lot)
CRUW [49]	Radar Tensor	2021	Object Detection	Radar (TI AWR1843, DCA1000), Cameras	Roads (parking, campus, city, highway)
RADDet [50]	Radar Tensor	2021	Object Detection	Radar (TI AWR1843), Stereo Cameras	Roads (urban, rural, highway, intersections, weather conditions)
Boreas [51]	Radar Tensor	2022	Object Detection, Localization, Odometry	Radar (Navtech CIR304-H), Camera, LiDAR	Roads (highway, rural, urban)
ColoRadar [52]	Radar Tensor, Point Cloud	2022	Localization	Radar (AWR1843), LiDAR, IMU	Indoor, outdoor environments
K-Radar [53]	Radar Tensor	2022	Object Detection, Object Tracking, SLAM	Radar (RETINA-4ST), Stereo Cameras, LiDAR	Roads (highway, intersection, urban)
OORD [54]	Radar Tensor	2024	Place Recognition	Radar (Navtech CTS350-X), GPS/INS	Roads (off-road, difficult terrain, naturalistic environments)
nuScenes [55]	Point Cloud, Grid Map	2019	Object Detection, Object Tracking	Radar (Continental ARS408), Camera, LiDAR	Roads (intersection, crosswalk, roundabout, pedestrian crossing)
Astyx [56]	Point Cloud	2019	Object Detection	Radar (Astyx 6455 HiRes), Camera, LiDAR	Roads (highway, urban, rural, parking, roundabout)
SeeingThroughFog [57]	Point Cloud	2020	Object Detection	Radar (77GHz), Stereo/Gated/FIR Cameras, LiDAR	Adverse road conditions (clear, rainy, snowy, foggy, nighttime, urban, highway, rural, traffic)
AIODrive [58]	Point Cloud	2020	Object Detection, Semantic Segmentation, Object Tracking, Depth Estimation	Radar (77GHz), RGB/Stereo Cameras, LiDAR	Roads (highway, residential street, parking)
RadarScenes [59]	Point Cloud	2021	Semantic Segmentation, Object Tracking	Radar (77GHz), Documentary Camera	Roads (urban, suburban, rural, highway, tunnel, intersection, roundabout, parking)
Pixset [60]	Point Cloud	2021	Object Detection, Object Tracking	Radar (TI AWR1843), Cameras, LiDARs	Roads (urban, suburban, highway)
VoD [61]	Point Cloud	2022	Object Detection, Object Tracking	Radar (ZF FRGen 21), Stereo Camera, LiDAR	Roads (highway, rural, urban)
TJ4DRadSet [62]	Point Cloud	2022	Object Detection, Object Tracking	Radar (Oculii Eagle), Camera, LiDAR	Roads (intersections, one-way streets)
aiMotive [63]	Point Cloud	2022	Object Detection	Radar (77GHz), Camera, LiDAR, GPS, IMU	Roads (highway, urban, rural)
WaterScenes [64]	Point Cloud	2023	Object Detection, Instance/Semantic/Waterline/Panoptic Segmentation	Radar (Oculii Eagle), Camera, GPS, IMU	Waterways (river, lake, canal, moat)
NTU4DRadLM [65]	Point Cloud	2023	SLAM	Radar (Oculii Eagle), RGB/Thermal Cameras, LiDAR, GPS, IMU	Roads (carpark, garden, campus)
Dual-Radar [66]	Point Cloud	2023	Object Detection, Object Tracking	Radar (ARS548 RDI, Arbe Phoenix), Camera, LiDAR	Roads (urban, suburban, highway, tunnel, parking)
MiliPoint [67]	Point Cloud	2024	Activity Recognition	Radar (TI IWR1843), Stereo Camera	Activities (identification, action classification and keypoint estimation)
V2X-Radar [68]	Point Cloud	2024	Object Detection	Radar (Oculii Eagle, Arbe Phoenix), Camera, LiDAR	Roads (various weather, time and intersection scenarios)
Dop-NET [69]	Micro-Doppler Signature	2020	Classification	Radar (Ancortek 24GHz)	Gestures (wave, pinch, click, swipe)
Open Radar Datasets [70]	Micro-Doppler Signature	2021	Classification	Radar (TI AWR2243), Camera, GPS, IMU	Roads (urban, highway, rural)
MCD-Gesture [71]	Micro-Doppler Signature	2022	Classification	Radar (TI AWR1843)	Gestures (push, pull, slide left, slide right, clockwise turning, counterclockwise turning)

1) *Datasets*: With the development of radar technology and increased computing capability, radar data characterized by ADC signal has emerged and garnered widespread attention in recent years. RaDICaL [41] is the first dataset that offers raw ADC signal data collected explicitly for road scenarios in autonomous driving. With access to raw radar measurements, the authors encouraged researchers to design novel processing methods to perform object detection directly or to get downstream data representations. RADIAL [42] is the most comprehensive dataset in terms of radar data representations. In addition to providing ADC signals, the RADIAL dataset contains processed data derived from the ADC signals, including radar tensors and point clouds. Apart from using ADC signals in road scenarios, another significant application is in-cabin vital sign monitoring. However, existing datasets in this area (e.g., [72], [73], [74], [75]) primarily focus on indoor scenarios like hospitals or home environments, which differ from in-cabin scenarios in autonomous driving. Although some studies (e.g., [11], [12], [76]) investigated monitoring occupant vital signs in cabins, their datasets are unavailable publicly.

2) *Methods*: The ADC signal representation can be applied for both surrounding traffic perception outside the vehicle and occupant vital sign monitoring inside the vehicle. Object perception on roads focuses on detecting vehicles and pedestrians using Doppler shifts and other radar features. Vital sign monitoring in a cabin emphasizes extracting and analyzing physiological information from radar ADC signals to monitor occupants. Since these two situations are handled differently, we classify methods using ADC signals in autonomous driving into road object-based and vital sign-based categories.

a) *Road Object-Based*: In recent years, there has been a growing interest in the development of road object perception using ADC signals, as evidenced by research in motion classification (e.g., [77], [78], [79], [80]) and object detection (e.g., [81], [82], [83], [84]). These studies encompass two main research directions. The first direction involves designing individual end-to-end learnable radar perception models (e.g., ADCNet [81], T-FFTRadNet [82], and CubeLearn [80]), enabling object perception on ADC signals. These approaches utilize deep learning on ADC signals as an alternative to traditional signal processing procedures. This solution replaces computationally intensive FFTs and simplifies data flow in embedded implementations, thus significantly reducing the computational requirements.

Another direction of road object-based methods focuses on improving the resolution of radar data, which is essential for estimating object locations and velocities. For example, the ADC Super-Resolution (ADC-SR) model [84] leverages ADC signals to enhance the radar's azimuth resolution, tackling complexities in ADC signals and predicting signals from unseen receivers. Consequently, the hallucinated ADC signals can further refine RAD tensors, enhancing object detection capabilities in autonomous driving.

b) *Vital Sign-Based*: The vital sign-based methods focus on monitoring the physiological signals of occupants inside the vehicle cabin, including heart rate, respiration rate, and presence detection [11], [76], [85], [86], [87]. Signal

processing techniques, such as spectrogram analysis, Short Time Fourier Transform (STFT) and Wavelet Transform (WT), are employed to extract the vital sign signals from the radar ADC signal [10], [87]. Machine learning algorithms, such as statistical classifiers or regression models, are trained using labeled data to estimate vital sign parameters from the extracted features [10].

Radar systems used for cabin vital sign monitoring are generally designed for close-range operation, such as within a vehicle or confined space. They focus on high-resolution detection of small movements and physiological changes, and use less power and have smaller form factors than road-oriented radar systems. However, vital sign-based methods using machine learning algorithms have limitations in real-time processing, which is crucial in low-latency applications [88], [89].

3) *Discussion*: Given that the ADC signal encompasses raw data derived from object reflections, it inherently carries the richest information about the object. This information can be effectively utilized for diverse tasks involving fine-grained feature recognition, such as activity classification, hand gesture recognition, and vital sign monitoring. However, there are certain limitations that should be acknowledged. One limitation is the sensitivity of the radar signals to factors such as electromagnetic interference, occupant positions and obstructing objects, requiring some advanced signal processing methods to extract features.

Another limitation is that ADC signals require a developer kit for data acquisition, which makes data processing challenging. Real-time object perception and vital sign monitoring pose computational challenges, requiring efficient hardware and software solutions. Moreover, neural networks processing ADC signals are suitable for classification tasks in closed environments, such as determining the presence or absence of objects and distinguishing between different types of objects [90]. However, in open environments, various distracting factors lead to inaccurate object feature identification.

B. Radar Tensor

As mentioned before, performing 3D FFTs on ADC signals along the sample, chirp, and antenna dimensions yields 3D RAD tensors. As is shown in Fig. 7, with these three features, two forms of radar tensors are formed: one is in 2D, including the Range-Azimuth (RA) tensor, Range-Doppler (RD) tensor and Azimuth-Doppler (AD) tensor; the other is the whole 3D RAD tensor, with each side consisting of a 2D tensor. Specifically, each 2D tensor represents a 2D pseudo-image that describes the spatial pattern of the received echo. The brighter colors within the tensor represent greater reflection amplitude at that location [91]. Moreover, in contrast to the 3D radar tensor that lacks height information, the 4D radar tensor expands the data representation by incorporating power measurements in four dimensions: range, Doppler velocity, azimuth angle, and elevation angle.

1) *Datasets*: Datasets representing radar data in tensor form can be categorized into three main groups:

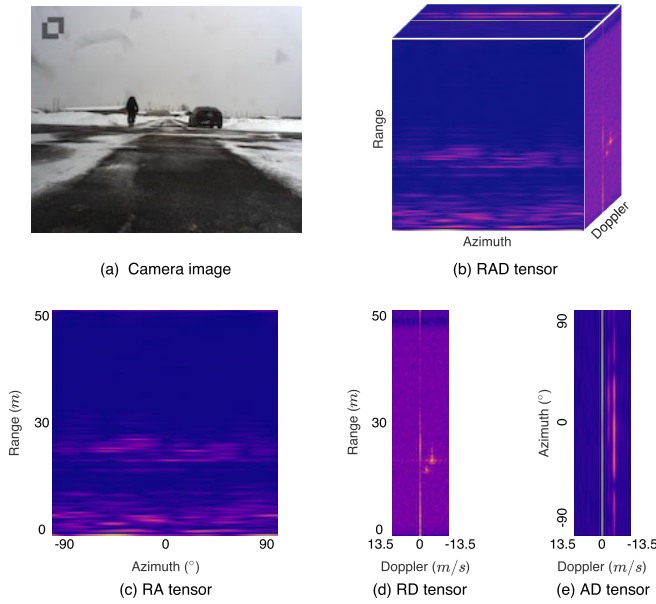


Fig. 7. Representation of radar tensor. (a) Image from the camera sensor. (b) 3D RAD tensor. (c) 2D RA tensor. (d) 2D RD tensor. (e) 2D AD tensor. Images are generated from the CARRADA [43] dataset.

- *2D tensors*: RADIATE [45], CRUW [49], FloW [92], MulRan [46], Oxford Radar RobotCar [47], SCORP [48], OORD [54];
- *3D tensors*: CARRADA [43], Zendar [44], RADDet [50], RADial [42];
- *4D tensors*: K-Radar [53].

The 2D radar tensor dataset has the largest number of all radar tensor datasets. Specifically, RADIATE [45], CRUW [49], MulRan [46], Oxford Radar RobotCar [47] and SCORP [48] are presented in 2D range-azimuth coordinates, representing the Bird's Eye View (BEV) position of objects. The FloW [92] dataset is in 2D range-Doppler coordinates, showcasing the relationship between range and Doppler velocity for each object. CARRADA [43] is the first dataset that combines synchronized stereo RGB images and 3D radar RAD tensors. It provides annotations with bounding boxes, sparse points, and dense masks for both range-Doppler and range-azimuth representations. Regarding 4D radar tensors, K-Radar [53] appears to be the only available dataset, offering comprehensive information on range, Doppler, azimuth, and elevation.

2) *Methods*: Radar tensors are expressed in various forms, with processing algorithms differing depending on tensor dimensions. Therefore, we categorize radar tensor methods into 2D, 3D, and 4D tensor-based methods based on currently available tensor dimensions.

a) *2D Tensor-Based*: 2D tensor-based methods typically process 2D input radar data by considering range-Doppler and range-azimuth information. To conduct object classification or detection on radar tensors, image-based network architectures (e.g., Faster R-CNN [93], ResNet [94], U-Net [95]) are applied or modified, as has been demonstrated in various studies using RA tensors [4], [91], [96], [97], [98], [99] or RD tensors [99], [100], [101], [102]. For example, Gao et al. [91] initiated the object detection process by defining a fixed-size

bounding box for each detected object on the RA tensor. In addition, they extracted the radar data within the bounding box from multiple RA tensors and arranged this data in the form of radar data cubes. With the temporal information and object movement pattern, the data cubes serve as features for classifying different objects, such as pedestrians, cars and cyclists. Dong et al. [96] applied ResNet [94] architecture on RA tensors for object detection. They also introduced uncertainty estimation for oriented bounding box localization further to enhance the accuracy of the object detection process. In general, 2D tensor-based methods employ a simplified radar data representation, reducing computational complexity. They can exploit correlations between two dimensions to improve detection and localization performance. However, these methods only employ part of the available 3D spatial information from the radar. With limited representation, the accuracy of complex scenarios with multiple objects or occlusions may be reduced.

b) *3D Tensor-Based*: 3D tensor-based methods process radar data simultaneously using the range, Doppler, and azimuth dimensions, forming a 3D RAD tensor. As such, this approach captures richer spatial information compared to 2D tensor-based methods, thus providing more valuable information to network architectures. Particular architectures have been developed to process aggregated views of 3D RAD tensors for object detection [3], [4], [50], [103], [104] and semantic segmentation [48], [105], [106], [107]. Specifically, Major et al. [3] first demonstrated the effectiveness of a deep learning-based object detection framework that operated on the RAD tensor and proved that the Doppler dimension helps increase detection performance. Similarly to the approach proposed by Major et al. [3], Gao et al. [103] decomposed the RAD tensor into three parts separately before combining them. The primary distinction between the two approaches is that the RA tensor in [103] consists of complex values that aid in recognizing objects using spatial patterns, thus increasing the accuracy of detection.

c) *4D Tensor-Based*: Recently, with the development of 4D radar sensors, methods on 4D radar tensors are emerging. K-Radar [53] compared the performance of 3D Range-Azimuth-Elevation (RAE) tensors with 2D RA tensors, demonstrating the importance of elevation information in radar perception. However, this method is still working on the 3D radar tensor without fully using the information of 4D radar tensors. Later, Enhanced K-Radar [108] improved K-Radar by introducing a new representation of 4D radar data named 4D Sparse Radar Tensor (4DSRT), which significantly reduces the 4D radar tensor size using offline density reduction. Specifically, 4DSRT retains the top-N% elements in Cartesian coordinates with the highest power measurements along the XYZ direction. The 3D sparse convolution [109] is then employed to extract the feature maps from the 4DSRT.

Compared to using 2D tensor, 3D tensor-based and 4D tensor-based methods offer a more comprehensive representation of radar data, leveraging the complete spatial information and allowing for precise object detection, segmentation, and localization. These methods can effectively handle complex scenarios involving multiple targets, partial occlusions, and

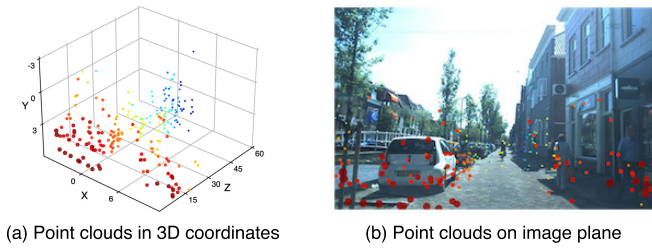


Fig. 8. Representation of point cloud. (a) Radar point clouds in 3D coordinates. (b) Point clouds projection on 2D image plane. Images are generated from the View-of-Delft [61] dataset.

varying azimuth angles. However, the increased dimensions of the tensor result in higher computational complexity and system delays.

3) *Discussion*: The radar tensor representation combines range, Doppler velocity, azimuth and elevation information into a coherent visual representation. This holistic view of the surrounding objects in the radar's field of view enables efficient perception and environment understanding. With these advantages, radar tensors are widely used in object detection and semantic segmentation tasks through combined labeling with cameras or LiDARs.

Although radar tensors retain more comprehensive information about an object, they also reserve noise and clutter information, which limit the ability to capture subtle object characteristics or distinguish between similar objects. Furthermore, the radar tensor provides a condensed representation of radar data, reducing the overall data size compared to raw ADC signals. Nevertheless, the radar tensor representation still requires considerable memory storage as well as a large bandwidth, especially for the new 4D radar tensors.

C. Point Cloud

By CFAR processing on the radar tensor, data in the format of a group of points, which we refer to as the point cloud, is obtained and visualized in Fig. 8(a). The point cloud provides a rough indication of an object's location, yet it cannot accurately describe the outline information of the object, as illustrated in Fig. 8(b). Each point contains information about the current reflection, including range, Doppler velocity, azimuth angle, and reflected power.

1) *Datasets*: Conventional 3D radar sensors generate sparse point clouds, as observed in datasets such as nuScenes [55], Zender [44], SeeingThroughFog [57], HawkEye [110], AIO-Drive [58], RADial [42], RadarScenes [59], aiMotive [63] and MiliPoint [67]. Recently, 4D radar datasets have emerged, including Astyx [111], VoD [61], TJ4DRadSet [62], WaterScenes [64], Dual-Radar [66] and V2X-Radar [68]. While Astyx [111] stands as the first 4D radar point cloud dataset, it is limited by a small data size of only 500 frames. VoD [61] and TJ4DRadSet [62] datasets represent notable advancements in terms of data categories and data size. Meanwhile, these two datasets also incorporate simultaneous LiDAR data, facilitating comparative analysis between 4D radar point clouds and LiDAR point clouds. Interestingly, WaterScenes [64] is a 4D radar dataset focused on objects in waterway environments,

which demonstrates the robustness of using 4D radar on water surfaces, particularly in adverse lighting and weather conditions. Dual-Radar [66] incorporates two types of 4D radars, expanding the comparison of different radar performance and further research on practical 4D radar perception algorithms. V2X-Radar [68] dataset integrates the on-vehicle 4D radar with the roadside 4D radar to achieve the Vehicle-to-Everything (V2X) cooperative perception in autonomous driving.

2) *Methods*: Point cloud-based radar algorithms take radar point clouds as input and are widely applied for object detection [5], [112], [113], [114], [115], semantic segmentation [23], [112], [116], [117], [118], [119], [120], [121], tracking [122], [123], [124], odometry [125], [126] and scene flow [127], [128], [129]. Based on various radar point cloud processing techniques, we categorize point cloud-based methods into three groups: clustering-based, point-based, and voxel-based methods.

a) *Clustering-Based*: Clustering-based methods [130], [131], [132] aim to group radar points into clusters using clustering algorithms (e.g., DBSCAN [40]), and then perform classification based on their spatial attributes. While clustering methods may be limited compared to deep learning-based methods, they are still effective in extracting individual objects from radar point clouds. The simplicity of clustering-based methods also reduces memory requirements, making them widely used in radar-based segmentation tasks. However, clustering-based methods are sensitive to parameter settings, such as the density threshold in DBSCAN. Additionally, these methods may also struggle with cluttered, overlapping, and closely spaced objects.

b) *Point-Based*: Point-based methods [5], [112], [113], [116], [117], [123], [133] process radar point clouds at the individual point level, extracting spatial features directly from these points. These methods generally draw inspiration from LiDAR-based algorithms, such as PointNet [134], PointNet++ [135] and Frustum PointNets [136]. These techniques leverage neural networks with convolutional filters to process radar points directly, considering spatial coordinates as integral features in the network structure. Schumann et al. [116] proposed a structure based on PointNet++ for semantic segmentation on radar point clouds. Specifically, the Multi-Scale Grouping (MSG) module in PointNet++ helps group and generate features for a center point and its neighborhoods. They demonstrated that incorporating the RCS value and compensated Doppler velocity significantly improved classification accuracy. Inspired by [116], Danzer et al. [112] adopted a two-stage method using PointNets [134], [136] for 2D car detection and segmentation. They considered each point as a proposal and adjusted the proposal size according to the object's prior knowledge. Then, PointNet and Frustum PointNets are used to classify the proposals and each point in the proposal. Finally, the bounding box prediction is executed only for the proposals that are objects. Point-based methods are good at accurate localization and classification of objects by considering individual radar points. However, they are susceptible to noise and occlusion, which affect feature extraction accuracy.

c) *Voxel-Based*: Voxel-based methods [114], [137], [138], [139] transform radar point clouds into volumetric representations by discretizing the 3D space into a grid of voxels. Then, 3D convolutions or sparse convolutional neural networks (e.g., VoxelNet [140], SECOND [141]) are leveraged to process the voxels to extract features for perception tasks. For example, GRIF Net [137] initially converts point clouds into voxels. As point clouds typically exhibit sparsity with numerous empty voxels, GRIF Net utilizes the Sparse Block Network (SBNNet) [142] to convolve only on masked regions, thereby avoiding unnecessary computations on ineffective blank areas. In LXL [139], the input radar point cloud is initially pillarized as in PointPillars [143]. Then, the pillar representation is fed into SECOND to extract multi-level BEV features with spatial and contextual information. Above all, voxel is an efficient representation for processing dense radar point clouds and is robust to varying point densities and occlusion. However, voxel-based methods lose fine-grained spatial information due to voxelization and discretization. Moreover, the limited resolution of voxel grids lead to decreased accuracy in object detection, especially for small or distant objects.

3) *Discussion*: Point cloud representation provides an intuitive 3D spatial structure of the surroundings. Since radar point clouds are generated by filtering techniques, they have advantages in filtering noise and being a lightweight data representation. However, some potential information within the raw data inevitably be lost [3]. Additionally, some small objects or objects with weak reflections may not be rendered as point clouds.

Currently, most point cloud detection and segmentation algorithms in radar are based on LiDAR algorithms. Point clouds in LiDARs are dense and can describe the outline of surrounding objects. Conversely, radar point clouds are incredibly sparse, posing challenges to developing practical algorithms. To address this issue, point cloud-based algorithms should explore the intrinsic relationships between point clouds, which can provide additional features, including velocities and RCS values.

D. Grid Map

Leveraging multi-frame point clouds, some researchers transform 3D radar reflections into a BEV pseudo-image called the grid map [116]. While the grid map is presented as aerial views, it is intended to illustrate the complexity and diversity of environments that autonomous vehicles are navigating. There are two primary types of grid maps commonly used in radar-based perception: one is the occupancy-based grid map [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], which represents the obstacles and free-space information; the other is the amplitude-based grid map [144], [145], which displays the reflected power values in that particular area. As shown in Fig. 9(b), an occupancy-based grid map has a clear outline, making it more feasible for outline-based detection tasks (e.g., parking spaces). An amplitude-based grid map, as indicated in Fig. 9(c), emphasizes the reflective characteristics of different objects, rendering it more appropriate for rural roads with a few distinct objects.

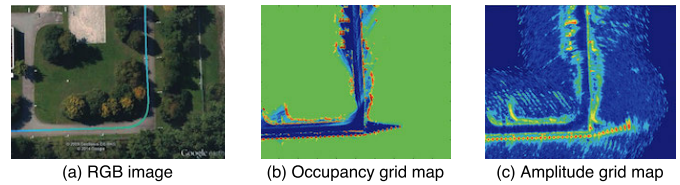


Fig. 9. Representation of grid map. (a) RGB image from BEV. (b) Occupancy grid map representing obstacles and free-space. (c) Amplitude grid map representing reflected power values. Images are from [144].

1) *Datasets*: There are two main datasets for radar grid map representation. Based on the nuScenes [55] dataset, Sless et al. [152] generated a grid map dataset for the semantic segmentation task, containing the actual occupancy state for each cell. Another grid map dataset is derived from Oxford Radar RobotCar [47], a radar dataset using a scanning radar without Doppler information for radar odometry tasks, utilizing the RA tensor as the data representation. Later, Qian et al. [154] extended this dataset to a grid map representation by creating 3D bounding boxes of vehicles from LiDAR point clouds. To facilitate grid-map object detection, they utilized visual odometry knowledge to synchronize the radar and LiDAR data.

2) *Methods*: As grid maps are generated from point clouds, clustering algorithms can be used to group radar measurements belonging to the same object. Besides, as the grid map has the same structure as the 2D image, deep learning-based algorithms can be directly implemented for detection and segmentation tasks. In general, methods for grid map representations can be categorized into two main types: occupancy-based and amplitude-based methods.

a) *Occupancy-Based*: Occupancy-based methods [144], [145], [146], [147], [148], [149], [150], [151], [152], [153], [155] utilize information such as the presence or absence of objects within a grid cell to make occupancy predictions. Traditionally, occupancy grid mapping is performed using an Inverse Sensor Model (ISM) and probabilistic methods (e.g., Bayesian filters [156]) to estimate occupancy probability [150], [157]. Additionally, Extended Kalman Filter (EKF) approaches take into account both measurement and motion models, modeling the environment as a Gaussian random field and estimating the state of each cell using a recursive Bayesian filter [158]. Occupancy-based methods provide a concise environmental representation, simplifying obstacle detection and localization for efficient path planning [159]. However, occupancy-based methods struggle to estimate occupancy states for cells with cluttered or overlapping radar returns [160]. Dynamic objects also pose problems due to their changing nature and intermittent radar measurements.

b) *Amplitude-Based*: Amplitude-based methods (e.g., [144], [145], [151], [161]) utilize the measured radar signal reflected intensity information for object detection and classification. Representative algorithms group radar measurements with similar amplitudes to identify potential obstacles. Clustering techniques (e.g., DBSCAN, K-means) are often used in this context [151]. In addition, convolutional neural networks are used as the classifier to distinguish between vehicles and non-vehicles in radar amplitude-based grid maps [145].

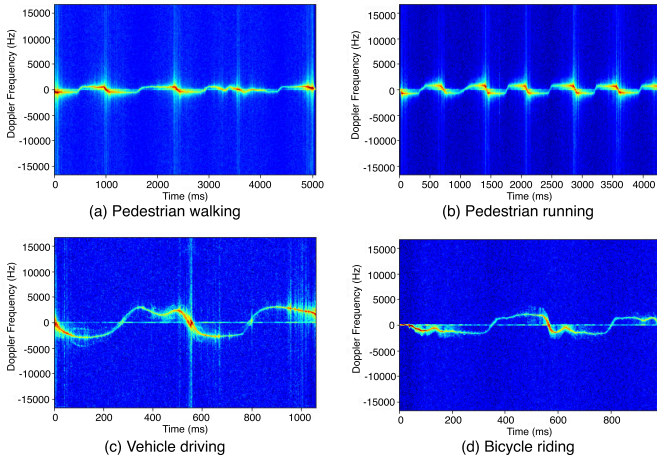


Fig. 10. Representation of micro-Doppler signature. (a) Spectrogram showing a pedestrian walking. (b) Spectrogram showing a pedestrian running. (c) Spectrogram showing a vehicle driving. (d) Spectrogram showing a bicycle riding. Images are generated from the Open Radar Datasets [70].

Amplitude-based methods offer robustness against clutter and noise, as they exploit variations in radar return amplitudes for object detection. However, amplitude-based representations can be sensitive to sensor noise and uncertainties, as they rely on accurate intensity measurements. The performance of amplitude-based methods is also influenced by factors like object occlusions, multi-path interference, and the radar's limited angular resolution.

3) *Discussion:* Radar grid maps provide spatial structure in BEV, offering interpretable and intuitive environmental understanding for geometric localization and obstacle detection. The grid map representation is useful for static objects, as the velocity information is ignored during map construction. By analyzing reflected power values, different object types (e.g., vehicles, pedestrians, buildings) can be distinguished, aiding semantic mapping and scene understanding.

However, as grid maps are constructed from radar point clouds, point cloud sparsity impacts detection, segmentation, and localization accuracy. This may reduce the accuracy of fine-grained object details or small objects in the grid map. Moreover, closely spaced objects are difficult to distinguish in radar grid maps.

E. Micro-Doppler Signature

Micro-Doppler signature refers to the representation of micro-motion, such as rotation and vibration caused by object parts, resulting in a characteristic representation that differs from the Doppler frequency variation. Micro-Doppler signature is generated by the Time-Frequency transform methods (e.g., STFT, WT) on Range-FFT results. As illustrated in Fig. 10, the spectrograms of walking and running pedestrians reveal different features. Specifically, the period of the micro-Doppler for a running pedestrian is shorter than that for a walking pedestrian. Moreover, the spectrograms between different types of objects (e.g., vehicles and bicycles) are unique, making them powerful features for object classification.

1) *Datasets:* The Open Radar Datasets [70] collects different types of outdoor moving targets, such as pedestrian walking and cycling, captured by a stationary radar system. The primary goal of this dataset is to utilize classification techniques for distinguishing between different motion activities. Apart from outdoor environments, some micro-Doppler signature-based datasets (e.g., Dop-NET [69], MCD-Gesture [71]) for motion recognition (e.g., health monitoring, gesture recognition) are from indoor environments. These indoor datasets can assist in occupant behavior recognition to some extent, but still differ from behavior classification in autonomous driving due to inconsistent acquisition environments.

2) *Methods:* Over the past few years, significant research attention has been paid to the classification and recognition of objects' postures and activities through micro-Doppler signature-based methods, particularly in the areas of object classification [91], [162], human activity and gait recognition [163], [164], [165], [166], and human gesture recognition [167]. Based on features of the micro-Doppler signature representation, we divide these methods into spectrogram-based methods and cadence velocity diagram-based methods.

a) *Spectrogram-Based:* Spectrograms are a commonly used representation for analyzing micro-Doppler signatures. Spectrogram-based methods simultaneously analyze the micro-Doppler signature in both the time and frequency domains, which has been applied to human motion recognition [164], [166]. The time-varying gait information, including the speed of arm and leg swing, is effectively encoded within the spectrogram representation. Thus, spectrogram-based methods provide feature localization of micro-Doppler signatures, allowing better identification and classification of different motions. However, spectrograms suffer from low time-frequency resolution trade-offs, leading to issues in capturing fast-changing Doppler signatures. Besides, spectrograms are also vulnerable to noise interference, which may obscure relevant information.

b) *Cadence Velocity Diagram-Based:* Apart from the spectrogram presenting Doppler information in the time domain, another representation of micro-Doppler signature is the Cadence-Velocity Diagram (CVD) that exists in the frequency domain, which is derived by performing a Fourier transform along the time axis of the spectrogram [168], [169]. It provides a metric for capturing the frequency repetition patterns ("cadence frequencies") over a given period. Thus, CVD-based methods provide a valuable measure for periodic changes in body parts at different velocities, such as walking or running [13], [165]. However, CVD-based methods become complex when dealing with targets having multiple moving parts or irregular motion patterns. Fine details or rapid changes in motion can be challenging to capture accurately, limiting the ability to distinguish between closely spaced components or rapidly changing motions [170].

3) *Discussion:* Micro-Doppler signature recognizes targets by observing minute movement characteristics of objects. This representation not only facilitates the differentiation of various object categories (e.g., pedestrians, bicycles, vehicles), but also

TABLE III
OVERVIEW OF DIFFERENT RADAR DATA REPRESENTATIONS

Representations	Main Tasks	Advantages	Limitations
ADC Signal	• Classification • Object Detection • Vital Sign Monitoring	• Contains all information from raw radar data • High temporal resolution • Fine-grained feature recognition	• Lack of semantic information • Complexity in data processing • Sensitive to noise
Radar Tensor	• Classification • Detection • Segmentation • Localization	• Provides spatial structure with range, azimuth and elevation • Provides rich motion information about the object	• Retains interference information around the object • Inconsistent resolution between range and azimuth
Point Cloud	• Detection • Segmentation • Tracking • Odometry	• Offers 3D spatial representation of the surrounding environment • Light-weight representation for objects • Filters weak reflected signals	• Sparse in describing the shape of objects • Susceptible to noise and occlusion • Loss of potential information after CFAR processing
Grid Map	• Mapping • Localization • Odometry • Detection	• Provides structured BEV representation of the surroundings • Offers a concise representation focusing on presence and occupancy	• Difficulty in representing complex and dynamic scenes • Suffers from noise, occlusions and resolution
Micro-Doppler Signature	• Motion Classification • Vital Sign Monitoring	• Contains information about object motion • Enhances differentiation between objects	• Limited resolution in range estimation • Vulnerable to noise and interference

enables the recognition of intricate object behaviors like gait and gesture recognition.

However, the transformation to the frequency domain via the Time-Frequency transform suffers from overlapping target components, resulting in suboptimal feature extraction for downstream tasks [78]. Similarly to ADC signal representation, the micro-Doppler signature is insufficient in delineating an object's spatial coordinates for detection and segmentation tasks, so it is commonly employed for classification tasks. Moreover, the micro-Doppler signature representation has limited resolution in range estimation compared to other representations like the radar tensor. This limitation could impact the ability to accurately determine the exact distance of an object, potentially affecting collision avoidance algorithms.

IV. CHALLENGES AND RESEARCH DIRECTIONS

In this section, we discuss challenges and potential research directions associated with radar perception in autonomous driving. When conducting academic research or application development on radar-based perception, the first challenge is what kind of radar product and radar data representation should be selected, as they all have their advantages and application scenarios. Subsequently, due to the inherent shortcomings of the radar sensor, such as sparsity, noise and limited resolution, we examine how to improve radar perception performance. Finally, in the trend of multi-sensor fusion with radar as an essential component in autonomous driving, we explore challenges in integrating radar perception with additional sensor modalities, including LiDARs and cameras. This integration aims to enhance the overall accuracy and robustness of perception by leveraging the complementary strengths of each sensor.

A. Which Data Representation Should Be Chosen?

As is described in Table III, different radar data representations have unique advantages and limitations, which should be considered when making informed choices. The choice of

data representation depends on the specific requirements of the autonomous driving system, including application scenarios and computational efficiency.

1) *Consideration of Application Scenarios:* Autonomous driving vehicles are categorized into six levels of autonomy by the SAE International framework, ranging from Level 0 (no automation) to Level 5 (full automation) [171]. At lower levels of autonomy (Levels 1 and 2), traditional 3D radar sensors are often sufficient for tasks such as adaptive cruise control and lane-keeping assistance, where limited spatial resolution is acceptable. However, as vehicles progress to higher levels of autonomy (Levels 3 and 4), the demand for more precise and robust sensor data increases, making 4D radars a critical component due to their ability to provide denser point clouds, higher resolution, and accurate object tracking. At Level 5, fully autonomous systems rely on a fusion of sensors, including 4D radars, LiDAR, and cameras, to achieve robust perception in complex and dynamic environments.

For applications requiring accurate object detection and tracking, representations like radar tensors or point clouds prove valuable. Radar tensors, such as RAD tensors, offer visualizations of range, Doppler velocity, and azimuth, which can be used to extract features using image-based methods [50], [106]. Point cloud representations deliver 3D spatial information aligned with real-world object locations, enabling utilization with point-based network architectures [172], [173]. Additionally, point clouds are the output format of conventional radar, benefiting from well-developed product suites and research materials [174].

When considering effective mapping and path planning, grid map representation emerges as a valuable choice. The Grid map offers a structured representation of surroundings, with occupancy grids providing a discretized layout. Grids can be classified as occupied or representing free space, aiding vehicle navigation and optimal path planning.

If the research focuses on vehicle cabins, radar representations of ADC signals and micro-Doppler signatures provide powerful insights. By analyzing the characteristics of the

TABLE IV
OVERVIEW OF CHALLENGES AND RESEARCH DIRECTIONS ON IMPROVING RADAR PERCEPTION PERFORMANCE

Topic	Challenges	Research Directions
Creating High-Quality Datasets	• Calibration • Annotation	• Develop adaptive and self-calibration algorithms • Design real-time and dynamic calibration techniques • Explore auto-labeling techniques
Dealing with Inherent Limitations	• Sparsity and limited resolution • Noise and clutter	• Explore the distribution of radar output • Construct features based on physical properties • Model inherent uncertainty
Improving Data Processing Algorithms	• Parameter estimation • Feature extraction	• Use neural networks instead of traditional FFTs • Apply graph neural network to extract features • Utilize occupancy prediction for scene understanding
Employing 4D Radar Sensors	• Large size • Perception algorithms	• Extract core values from tensor dimensions • Develop efficient 3D object perception models • Design specific architectures on dense information

ADC signal, such as its amplitude, frequency, and temporal variations, it becomes possible to monitor vital signs. Additionally, micro-Doppler signatures extract detailed information about human movements, differentiating between vibrations caused by respiratory movements, heartbeats, or even gestures unrelated to vital signs. Thus, these two representations can be utilized in the cabin for health monitoring, detecting distress, and alerting drivers to emergency services.

2) *Consideration of Computational Efficiency*: If real-time processing and low computational complexity are critical, computationally efficient representations such as radar tensors and point clouds are preferable. The radar tensor represents radar echoes in a 2D or 3D matrix, while the point cloud representation is a set of individual 3D points representing detected objects. As far as computational performance is concerned, some tensor-based methods and point cloud-based methods have already achieved real-time object detection and segmentation [4], [23], [175], [176].

Conversely, ADC signals or 4D radar tensor provide radar data with high temporal resolution, but are typically large and computationally expensive to process in real-time, requiring additional algorithms for signal processing and feature extraction [18], [84]. As mentioned in RADIAL [42], the 4D radar tensor at each timestamp amounts to 450 MB, with a processing requirement of 45 GFLOPS for a single elevation, which increases to 495 GFLOPS when all 11 elevation are covered. On the other hand, the point cloud approach requires only 8 GFLOPS to compute the 3D coordinates for a sparse cloud of about 1000 points. In comparison, the processing of a 256-pixel image consumes only 0.4 GFLOPS of computational resources.

Micro-Doppler signatures require additional processing, but their feasibility for real-time depends on specific algorithms and computational resources [177], [178]. Grid maps are structured grid-based formats, simplifying subsequent processing and potentially enabling computational efficiency. However, they may not meet real-time requirements if frequent updates are needed for new scenarios [158], [179].

B. How to Improve Radar Perception Performance?

The constraints inherent in radar sensors and the limitations associated with radar data processing algorithms present

challenges to the performance of radar perception. Improving radar perception performance is crucial for autonomous driving, enabling more accurate decisions and predictions. In the following, we explore four areas: creating high-quality datasets, dealing with inherent limitations, improving data processing algorithms, and employing 4D radar sensors. An overview of challenges and research directions in these four areas is described in Table IV.

1) *Creating High-Quality Datasets*: A high-quality dataset provides more accurate ground truth information, enabling the algorithm to learn the correct associations and features required for accurate object classification, detection, tracking, and localization. However, creating a high-quality dataset is challenging for radar, primarily in terms of calibration and annotation.

a) *Calibration*: Ensuring accurate calibration for radar sensors is a prerequisite for radar perception. The triangular corner reflector is a common target for radar calibration. When radar waves encounter the corner reflector, they bounce off each surface and converge at a point, forming a strong reflection that the radar sensor can easily measure [180]. While significant progress has been made in radar calibration techniques (e.g., [181], [182], [183]), there are still challenges to address and potential research directions to explore.

Designing calibration methods adapted to varying environmental conditions is essential. Research efforts should focus on developing adaptive and self-calibration algorithms that account for changes in clutter, interference, and propagation conditions [184]. Developing advanced algorithms that handle complex system configurations is a promising research direction. Techniques such as machine learning, optimization, and statistical approaches can be explored to improve calibration accuracy and efficiency [185]. In addition, dynamic radar calibration during driving is challenging. Developing dynamic calibration techniques, such as the Gaussian Mixture Model (GMM), that adapt to changing systems and environments enhances the operational reliability of radar systems [185]. Addressing these challenges and exploring these research directions can lead to improved radar calibration techniques, enabling more accurate and reliable radar measurements in autonomous driving.

b) Annotation: In another aspect, high-quality datasets with accurate and rich annotations are crucial for training and validating perception algorithms. However, the process of data annotation is labor-intensive and time-consuming. This is particularly true when dealing with radar data, as the inherent representation of objects within this data modality does not fully reveal their physical forms. Data describing objects with ground truth information, such as position, velocity, reflected power, and class, should also be annotated to enable training models that generalize well to real-world conditions.

The exploration of auto-labeling techniques for radar data presents a promising avenue to mitigate the burden of manual data labeling. In practice, radar data labels can be derived by leveraging corresponding ground truth from camera images and the extrinsic matrix of radar and camera sensors [64], [181]. However, the efficacy of this labeling approach for radar data remains an issue, given that radar targets may not consistently align with the ground truth depicted in images. Thus, despite the potential benefits associated with auto-labeling radar data, effectively filtering extraneous data around objects remains an ongoing challenge.

2) Dealing With Inherent Limitations: While radar sensors offer significant advantages for perception in autonomous driving, they also have certain limitations. Factors such as adverse weather conditions, interference, and the trade-offs between range, resolution, and field of view are significant considerations that impact the reliability and effectiveness of radar systems. Understanding these limitations is crucial for designing appropriate strategies to create a robust and comprehensive perception system.

a) Sparsity and Limited Resolution: The sparse nature of radar data provides less detailed information about object shape and fine structure than LiDAR and camera data. Limited resolution hinders the radar system's ability to distinguish between closely spaced targets, leading to potential confusion and misinterpretation. Consequently, radar sensors struggle to provide precise localization or detailed shape information, making it challenging to classify and discern fine-grained object features.

The frequency of radar systems plays a critical role in determining data representation parameters, particularly resolution and the ability to discriminate between surfaces and objects with varying textures or roughness. Modern automotive radars, which operate at higher frequencies such as 77 GHz or 79 GHz, benefit from increased bandwidth, leading to improved range resolution. This allows for better differentiation between closely spaced objects and enhances the detection of subtle variations in distance. In addition to spatial resolution, higher frequencies enhance the radar's ability to detect variations in surface textures and material properties, aiding in object classification and recognition. However, these advantages come with trade-offs, such as increased signal attenuation due to atmospheric absorption and reduced penetration through certain materials, which must be carefully considered in system design.

While combining multiple frames enhances accuracy, it can also introduce system delays [61], [116], [137], [186], [187], [188], [189]. Column or pillar expansion methods (e.g., [186],

[187], [190]) complement the lack of measurement in the vertical direction, but only alleviate ambiguity to a limited extent. Exploring radar output distribution (e.g., Gaussian distribution) based on the fundamental principles of radar presents a valuable research opportunity [191]. Moreover, existing methods (e.g., PointNet [134], PointNet++ [135]) designed for processing dense LiDAR point clouds are limited for handling sparse radar point clouds [192]. Therefore, developing specialized architectures to address the challenges of sparse radar detection presents a promising research direction. For example, Liu et al. [23] leveraged cosine similarity loss and normalized inner product loss as part of the training process for sparse radar detection points, enhancing the performance of point-wise center shift vector-guided clustering.

While conventional radar systems with a limited number of antennas may exhibit constrained angular resolution, this limitation can be effectively addressed through Synthetic Aperture Radar (SAR) techniques. By exploiting the motion of the vehicle, SAR can synthesize a large virtual aperture through advanced signal processing techniques, thereby enhancing angular resolution. This approach has been extensively studied in recent works [193], [194], [195], [196], [197], [198], [199], demonstrating its feasibility and effectiveness in automotive radar applications. For instance, [193] and [194] explored cooperative imaging in multi-sensor networks, particularly addressing phase synchronization and data processing to achieve high-resolution imaging. [195], [196], [197] analyzed the potential and challenges of automotive SAR imaging, proposing motion estimation and compensation techniques and navigation-aided imaging methods to mitigate the impact of vehicle motion on imaging quality. [198] and [199] analyzed the impact of urban environments on SAR imaging, proposing multi-beam SAR imaging techniques to address multi-path effects and occlusion issues in complex urban settings.

Given SAR's ability to achieve spatial resolution comparable to LiDAR, the representation and storage of SAR data require careful consideration. High-resolution SAR data can be represented as point clouds, range-Doppler maps, or grid maps, with point clouds being particularly suitable for applications like autonomous driving due to their compatibility with LiDAR-based systems. However, the increased data volume necessitates efficient storage solutions, such as compression algorithms and hierarchical data structures. The choice of output format depends on the specific application, with point clouds being preferred for environment mapping and range-Doppler maps for tasks like target detection.

b) Noise and Clutter: Radar sensors are susceptible to noise, resulting in false detections and inaccurate object positions. Noise arises from various sources, including electronic components within the radar system, thermal noise, atmospheric disturbances, and electromagnetic interference. Clutter refers to unwanted radar returns caused by environmental factors such as ground reflections, vegetation, or other structures.

The signal received by the in-cabin radar sensor not only contains information about the occupant's vital signs, but also includes unwanted motion induced by engine vibrations [11]. Consequently, monitoring vital signs in a non-stationary

environment using a radar sensor becomes rather challenging. Traditional techniques (e.g., [200], [201], [202], [203]) applied to reduce noise in automotive radars usually draw upon CFAR and peak detection algorithms. However, these methods have shown limitations in terms of adaptability when faced with varying conditions and unpredictable noise types.

For noise removal in radar tensors, researchers draw inspiration from principles and techniques used in image denoising and restoration [204], [205], [206], [207], [208], [209], [210]. By considering radar tensors as images, various techniques, including convolutional neural networks, restoration filters, and wavelet-based methods, can be applied to reduce noise and enhance the quality of these tensors [211], [212]. For mitigating noise in point clouds, removing radar detections that fall outside boundaries defined by ground truth information from cameras or LiDARs is an approach [186], [213]. Constructing features based on physical properties can provide the necessary guidance for models to differentiate targets from clutter. This involves examining point density, spatial arrangement, and other statistical properties to help distinguish signal from noise. However, a unique challenge presented by radar point clouds is their inherent sparsity and inaccuracy. Developing techniques to account for this uncertainty could allow for more robust noise removal and aid in distinguishing objects from noise.

3) *Improving Data Processing Algorithms*: Processing raw radar data to extract meaningful object information is critical for perception accuracy. Advanced signal processing techniques suppress clutter and noise in radar measurements, including adaptive beamforming and spatial filtering [214], [215]. However, data processing using these signal processing techniques remains challenging for radar perception. Utilizing machine learning and deep learning algorithms can also improve perception and object recognition capabilities. We focus on discussing two aspects of algorithm improvement: parameter estimation and feature extraction.

a) *Parameter Estimation*: Radar signals often exhibit complex patterns and non-linear dependencies. Traditional FFT-based methods only capture part of the relevant information required for parameter estimation [4], [216]. One potential research opportunity is employing neural networks to extract radar parameters instead of traditional FFT operations. This approach is valuable to reduce computational requirements, while simplifying the data flow within embedded implementations [4], [217], [218]. An example can be found in RODNet [4], where FFT operations are selectively applied in the sample and antenna dimensions, while the chirp dimension is preserved to form the range-azimuth-chirp tensor. Then, a neural network is employed to process the chirp dimension for extracting relevant Doppler features. This end-to-end learning approach allows the neural network to learn relevant features and patterns from the radar data, potentially improving the accuracy and efficiency of parameter extraction.

b) *Feature Extraction*: Graph Neural Networks (GNNs) [219] present a promising research direction, enabling operations on graph data to effectively capture relationships between elements within complex structured data (e.g., point clouds, images) [219]. Radar data can also be represented

as a graph, where each graph node represents a point from point clouds or a pixel from radar tensors, and the edges capture the spatial or temporal relationships between them. Then, Graph Convolutional Networks (GCNs) employ graph convolutional layers to update node embeddings by aggregating and transforming information from neighboring nodes. These layers effectively capture spatial dependencies between objects, facilitating feature learning in graph-based radar data representations. GNN-based methods applied to radar point clouds (e.g., Radar-PointGNN [220], RadarGNN [221]) or radar tensors (e.g., GTR-Net [222]) demonstrate graph representations' effectiveness in capturing contextual and spatial information, thereby improving the overall performance.

Recently, occupancy prediction (e.g., SurroundOcc [223], OpenOcc [224], Occ3D [225], TPVFormer [226]) using cameras and LiDARs has been a hot research topic in autonomous driving. Compared to 3D bounding boxes, 3D occupancy can not only describe target objects, but also indicate background state and capture fine-grained details [224]. Radar occupancy prediction presents a valuable research direction for radar perception in autonomous driving. By analyzing occupancy patterns in radar data, the structure of the scene, including road boundaries as well as obstacles, can be inferred. Potential research could explore using contextual semantics to enhance radar occupancy prediction accuracy, enabling more comprehensive environment understanding.

4) *Employing 4D Radar Sensors*: Radar sensors have made significant progress, transitioning from 3D to 4D capabilities, including enhanced resolution and elevation measurement capabilities. As a result, more and more research is focused on the 4D radar, with growing numbers of radar datasets and algorithms proposed recently. Given these advancements, employing 4D radar sensors is a promising and potential research direction for radar perception in the broader domain of autonomous driving. However, 4D radar is still challenging in terms of large data size as well as perception algorithms.

a) *Large Size*: The spatial distribution of objects is effectively captured and represented within 4D radar datasets, such as VoD [61], TJ4DRadSet [62], K-Radar [53], Water-Scenes [64], and Dual-Radar [66]. However, compared to 3D radar sensors, 4D radars produce prohibitively large data outputs. For example, in the K-Radar [53] dataset, the size of 4D radar tensor only in the forward direction amounts to 12TB, while the size of the synchronously acquired 360-degree LiDAR point cloud data is only 0.6TB. Thus, methods of size reduction are worth consideration. Preliminary investigations involve developing Enhanced K-Radar [108], which extracts higher value by sampling different tensor dimensions. This innovative approach shows potential for enhancing training speed and reducing memory requirements.

b) *Perception Algorithms*: Emerging perception algorithms applied to 4D radar datasets also indicate that the 4D radar data significantly enhances radar perception capabilities [61], [114], [173], [227], [228]. With various architectures (e.g., point-based, voxel-based, pillar-based) proposed, Palmer et al. [229] conducted a comprehensive analysis of detection performance achieved by existing models on 4D radar datasets, including VoD [61] and Astyx [111].

TABLE V
OVERVIEW OF CHALLENGES AND RESEARCH DIRECTIONS ON RADAR DATA FUSION

Topic	Challenges	Research Directions
Associating Heterogeneous Data	- Projection-based or geometric-based associations result in ineffective alignment - Sensitive to occluded radar returns and background clutter	- Utilize BEV features to enhance spatial understanding - Apply attention mechanisms to mitigate background clutter - Employ joint probabilistic data association methods to estimate the likelihood of associations
Handling Challenging Conditions	- Sensitive to adverse weather and object occlusions - Uncertainty estimation of different sensor measurements	- Develop adaptive fusion strategies to adjust fusion parameters based on environmental conditions - Employ probabilistic fusion techniques to model and propagate uncertainties

They evaluated the performance of Voxel R-CNN [230], SECOND [141], PointRCNN [231], and PV-RCNN [232] through cross-model validation and cross-dataset validation experiments. Numerical results showed no clear best model, with the performance varying by object class and distance. Further research is expected to develop efficient 3D object detection models based on 4D radar data, utilizing rich data and feature extraction to address challenges.

C. How Can Data Fusion Be Applied to Different Radar Representations?

While radar sensors are good at detecting and ranging objects, they provide limited information about object characteristics like color, texture, and fine-grained visual details. Integrating radars with complementary sensors, such as cameras, LiDARs and ultrasonic sensors, overcomes this limitation and provides comprehensive perception solutions [1], [233], [234], [235], [236]. Researchers can leverage the complementary strengths of each modality to achieve a balance between performance, cost, and robustness. Sensor fusion techniques further enhance system reliability by combining range and velocity data from radar with high-resolution spatial data from other sensors.

When fusing radar from multiple sensors, the choice of representation depends on the specific application requirements, the type of radar systems involved, and the desired level of information granularity. Common approaches to multi-radar fusion can be categorized into three levels: data-level fusion, feature-level fusion, and object-level fusion [1]. Data-level fusion involves combining the unprocessed radar signals from multiple radars. This approach preserves the maximum amount of information, as it avoids any loss of detail during intermediate processing steps. Feature-level fusion extracts meaningful features from the raw data of each radar system, such as target range, velocity, Doppler shift, and angular position. These features are then combined to form a unified representation of the detected targets. This approach strikes a balance between computational efficiency and information preservation, making it a popular choice for many practical applications. Object-level fusion involves processing the raw data from each radar system independently to make local decisions (e.g., object detection, classification, or tracking) and then combining these decisions to produce a final result. Common methods for object-level fusion include majority voting, weighted averaging, and Bayesian inference. However,

radar data fusion also faces some challenges, particularly in heterogeneous data association and in challenging condition processing. An overview of these challenges and research directions is summarized in Table V.

1) *Associating Heterogeneous Data*: A significant challenge is the ambiguity in associating radar data with other modalities, as they are heterogeneous. For radar-camera fusion, existing approaches involve projecting radar data onto the image plane and subsequently establishing their correspondence via a calibration matrix [16], [111], [186], [237], [238]. For radar-LiDAR fusion, existing methods exploit geometric information to associate radar and LiDAR point clouds by comparing the position, velocity, and shape of the detected objects to determine if they correspond to the same physical object [239]. However, projection-based or geometric-based associations result in ineffective alignment with object centers. As previously stated, radar data exhibits sparsity, inaccuracy, and noise, thereby causing inadequate associations in both object-level and data-level fusion scenarios.

Therefore, associating radar data with other modalities is a critical but challenging question. In our opinion, the incorporation of BEV features, transformer architectures, and attention mechanisms are valuable research directions that can significantly improve fusion performance. BEV provides a top-down perspective view of the vehicle's surroundings, thereby enhancing spatial understanding. Using BEV transformations and transformer architectures, radar-camera fusion methods (e.g., RCBEV [240], CRAFT [241], CRN [242], RCFusion [243], LXL [139]) or radar-LiDAR fusion methods (e.g., Bi-LRFusion [244], BEVGuide [245], ACF-Net [246], ST-MVDNet++ [247]) have proven to deliver impressive performance.

Moreover, attention mechanisms have been successfully utilized to deal with complex contextual relationships, which makes them promising for data fusion tasks [1]. Attention mechanisms play a crucial role in determining the relevance of different sensor modalities at different spatial or temporal locations. In the case of data association, the attention mechanism filters occluded radar returns and mitigates background clutter, which facilitates the alignment and matching of objects or features across the different modalities [188], [241]. Thus, attention-based association with adaptive thresholds emerges as a prospective approach for establishing a connection between radar and other modalities. Furthermore, the Joint Probabilistic Data Association (JPDA) method is a potential research direction that leverages Bayesian filtering techniques

to estimate the likelihood of associations based on the joint probability distribution of radar and other modalities [248]. By considering uncertainties associated with the measurements and estimates, JDPA enables robust data association.

2) *Handling Challenging Conditions*: Fusion in challenging conditions refers to the ability of a sensor fusion system to maintain reliable and accurate perception even in adverse environments, where individual sensors may face difficulties. In such conditions, sensor measurements are noisy, degraded, or influenced by poor visibility, adverse weather and object occlusions [249], [250], [251], [252]. Therefore, sensor fusion aims to leverage the complementary strengths of different sensors to compensate for the limitations of one sensor with the capabilities of another. Developing robust fusion algorithms that handle varying environmental conditions and adaptively combine sensor data is critical for reliable perception.

Fusion in challenging conditions requires adaptive fusion strategies that dynamically adjust fusion parameters based on environmental conditions. Developing data weighting techniques that assign appropriate weights to sensor measurements based on reliability can enhance fusion performance under challenging conditions [253], [254]. For example, radar sensors may be more reliable than cameras and LiDARs under adverse weather conditions. Therefore, the fusion algorithm should adaptively emphasize radar data and de-emphasize other sensors.

Effective fusion in challenging conditions also requires uncertainty estimation associated with sensor measurements. It refers to the lack of complete confidence or knowledge about the observations obtained from different sensors in the fusion process. Probabilistic fusion techniques should be incorporated to model and propagate uncertainties properly. Bayesian neural networks are notable techniques employed for uncertainty estimation, using the prior distribution of network weights to infer the posterior distribution. This enables the computation of probabilities for specific predictions, thereby enhancing the reliability of the fusion network [233], [255]. Such techniques should enable fusion algorithms to make informed decisions considering the uncertainty levels in fused data.

V. CONCLUSION

In conclusion, this review explores five radar data representations (i.e., ADC signal, radar tensor, point cloud, grid map, and micro-Doppler signature) in autonomous driving. Through an in-depth study of radar operating principles and signal processing, we reveal the generation process, applications, as well as the advantages and limitations of these representations, providing valuable insights for the continued development of autonomous driving technology. By investigating advanced methods for different data representations, we aim to gain insights into the characteristics of different algorithms as well as emerging trends for radar perception.

Through analyzing various radar data representations, we thoroughly discuss the key challenges and propose potential research directions for radar perception. In general, radar perception is progressing towards data representations containing rich information. On the one hand, representations

like ADC signals and radar tensors offer increased potential information, thereby holding significant value for radar perception. On the other hand, new 4D radar sensors bring forth denser point clouds and higher resolutions, representing a noteworthy trend in autonomous driving. In terms of radar perception networks, the incorporation of transformer architectures, attention mechanisms, and BEV features provides valuable research directions to improve the perception performance. Moreover, emerging radar-based sensor fusion works are also hot topics in autonomous driving and are expected in future works. Above all, we hope our review serves as a valuable reference for both researchers and practitioners in developing robust radar perception, making our vehicles and transportation systems safer and more efficient.

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active DT via immersive realities, path planning algorithmic advancement, and cost implication.



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